

CV — JAIME AGUILAR GUERRERO, Ph.D.

Research Associate, Physical Sciences Department

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PROFILE Jaime Aguilar Guerrero, Ph.D., is an early-career atmospheric and space physicist whose work centers on the observation, modeling, and visualization of gravity waves and upper atmospheric phenomena. Drawing on a solid foundation in remote sensing, numerical modeling, and optical imaging, he has both led and collaborated on multi-institutional research using advanced instrumentation and innovative data analysis. Jaime's passion for bridging scientific discovery, data interpretation, and public engagement is reflected in his development of 3D and mixed-reality visualization tools for Earth sciences. Active in the scientific community, he serves as a research associate, educator, and frequent presenter and convener at conferences including CEDAR, AGU, and EGU, and contributes as a poster judge. Jaime is dedicated to advancing knowledge in atmospheric dynamics, developing visualization tools for scientific research, and building collaborations across fields such as aeronomy, space physics, human-computer interaction, and computer vision.

PRIMARY RESEARCH AREAS

- Atmospheric Gravity Waves & Mesospheric Dynamics
- Remote Sensing & Optical Imaging
- Aeronomy & Space Physics
- Scientific Visualization & Data Analysis
- Computer Vision & 3D Modeling
- Virtual Reality & Spatial Computing in Geophysical Sciences and STEAM

EDUCATION *Ph.D., Engineering Physics*
Embry-Riddle Aeronautical University, Daytona Beach, FL, 2021
Focus: Space Physics and Aeronomy
Dissertation: "Acoustic/Gravity Wave Phenomena in Wide-Field Imaging" (Advisor: Jonathan B. Snively, PhD)

M.Sc., Engineering Physics
Embry-Riddle Aeronautical University, Daytona Beach, FL, 2015
Focus: Space Physics and Aeronomy
Thesis: "Climatology of Mesospheric Mountain Waves over Cerro Pachón, Chile" (Advisor: Alan Z. Liu, PhD)

B.Sc., Engineering Physics
ITESM, Monterrey, NL, Mexico, 2011
Research focus: Aeronautics, Optics

ACADEMIC & RESEARCH POSITIONS	<i>Embry-Riddle Aeronautical University, Daytona Beach, FL</i>	
	– Research Associate	2023 – Present
	– Postdoctoral Research Assistant	2022 – 2023
	– Research Assistant	2015 – 2021
	– Teacher Assistant	2012 – 2015

RESEARCH/ PROJECTS	<i>ONR's MURI – MITIGATE</i>	2026
	Project Manager	
	<i>NSF's CGWaveS Campaign</i>	2025 – Present
	Data analysis, Modeling	
		2024 – 2025
	<i>DARPA's AtmoSense/AIRWaveS</i>	
	Project Manager	2023 – Present
	<i>NASA's Atmospheric Waves Experiment (AWE)</i>	
ERAU's PI and Student Collaboration Lead	2025 – Present	
	<i>NASA's Vorticity Experiment (VortEx)</i>	
Atmospheric Modeling	2023 – Present	
	<i>AGU's EOS Article (in review): Show and Tell – Virtual Reality in Earth and Space Sciences</i>	
Coauthor		

PROFESSIONAL SERVICE	<i>Judge, Outstanding Student Poster and PICO (OSPP) Award, EGU General Assembly, 2025</i>	2025
	<i>Judge, Outstanding Student Presentation Award (OSPA), AGU Fall Meeting, 2023 & 2024</i>	2023, 2024
	<i>Judge, Student Poster Competition, CEDAR Workshop, 2023 – 2025</i>	2023 – 2024

TECHNICAL SKILLS SOFTWARE: MATLAB, Fortran, C++, JavaScript, Xcode, Blender, Creative Cloud Suite
LANGUAGES: English, Spanish, French (intermediate), German (basic)

JOURNAL PUBLICATIONS

Inchin, P. A., Kaneko, Y., Gabriel, A. A., Ulrich, T., Martire, L., Komjathy, A., ... & Snively, J. B. (2024). Three-dimensional numerical modeling of coseismic atmospheric dynamics and ionospheric responses in slant total electron content observations. *Geophysical Research Letters*, 51(21), e2023GL108068.

Inchin, P. A., Snively, J. B., Williamson, A., Melgar, D., Aguilar Guerrero, J., & Zettergren, M. D. (2020). Mesopause Airglow Disturbances Driven by Nonlinear Infrasonic Acoustic Waves Generated by Large Earthquakes. *Journal of Geophysical Research: Space Physics*, 125(6). doi:10.1029/2019ja027628.

Inchin, P. A., Guerrero, J. A., Snively, J. B., & Kaneko, Y. (2022). Simulation of Infrasonic Acoustic Wave Imprints on Airglow Layers During the 2016 M7. 8 Kaikoura Earthquake. *Journal of Geophysical Research: Space Physics*, 127(3), e2021JA029529. doi:10.1029/2021ja029529

CONFERENCE PRESENTATIONS & ABSTRACTS

(selected; includes
convener, invited
talks and posters)

Smith, B., Aguilar Guerrero, J., Boghosian, A., Collins, K., Lonjou, V., Rogers, L., Yang, C.P. Exploring Earth System Complexity with Digital Twins and 3D Visualization and Sonification II Oral. in AGU25.

Aguilar Guerrero, J., Bergsson, B., Mesfin, S., Inchin, P., Zettergren, M., Scherliess, L., Zhao, Y., and Pautet, D.: Observing Mesospheric Gravity Waves with NASA's AWE Mission and Correlating to GNSS TEC Maps, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-13773, <https://doi.org/10.5194/egusphere-egu25-13773>, 2025.

Snively, J. B., Calhoun, D., Aiton, S., Burstedde, C., Brandt, H., Griesbach, T., Pineyro, B., Mesfin, S., Aguilar Guerrero, J., Heale, C. J., Inchin, P., Sabatini, R., Hirsch, M., & Zettergren, M. D. (2024, December). Scalable Modeling of Acoustic-Gravity Wave-driven Traveling Ionospheric Disturbances. AGU Fall Meeting 2024, Washington, D.C., Session: SPA-Aeronomy / Traveling Ionospheric Disturbances Driven from Above and from Below I Oral, id. SA23D-03.

Aguilar Guerrero, J., Heale, C. J., Snively, J. B. (2023, December). A comprehensive simulated imaging analysis of a gravity wave event with ground-based instruments and brightness-weighted-temperature swaths from space borne imaging instruments. AGU Fall Meeting 2023, San Francisco, CA, Session: Atmospheric Sciences / Atmospheric Gravity Waves, from the Surface to the Edge of Space II Poster, Poster No. 2381, id. A53K-2381.

Aguilar Guerrero, J., Snively, J. B., Heale, C. J. (2024, December). Advancements in Volumetric Atmospheric Data Exploration Using Virtual Reality: A Case Study with the MAGIC Model. AGU Fall Meeting 2024, Washington, D.C., Session: Informatics / Planet Building: Advances in Data Visualization Methods and VR/AR/MR Spatial Interface Systems to Support Earth and Space Science I Poster, Poster No. 2134, id. IN11C-2134.

Snively, J. B., Calhoun, D., Aiton, S., Burstedde, C., Brandt, H., Hirsch, M., Aguilar Guerrero, J., Heale, C. J., Inchin, P., Mesfin, S., Pineyro, B., Sabatini, R., & Zettergren, M. D. (2023, December). Scalable Modeling of Acoustic-Gravity Wave Processes in the Ionosphere, Thermosphere, and Mesosphere (ITM). AGU Fall Meeting 2023, San Francisco, CA, Session: SPA-Aeronomy / Dynamics of the Thermosphere and Ionosphere System at Low and Middle Latitudes During Varying Internal and External Conditions III Poster, Poster No. 271, id. SA23B-271.

Aguilar Guerrero, J. (2024, December). Engaging the Public in Heliophysics: Deploying Atmospheric Data to AR and VR Platforms for Education and Outreach. AGU Fall Meeting 2024, Washington, D.C., Session: SPA-Aeronomy / The MacGyver Session: The Place for Novel, Exciting, Self-Made, Hacked, or Improved Sensors and Software Solutions for the Year of Open Science and the Heliophysics Big Year I Oral, id. SA11A-05. *Convener*.

Aguilar Guerrero, J., Heale, C. J., Inchin, P., Snively, J. B., & Zettergren, M. D. (2023, December). Visualization of volumetric atmospheric data with cutting-edge mixed reality platforms for both public dissemination and improving scientific analyses. AGU Fall Meeting 2023, San Francisco, CA, Session: Education / Amazing Technologies and Capabilities That Contribute to STEAM II Oral, id. ED34A-07.

Aguilar Guerrero, J., Heale, C. J., & Snively, J. B. (2023, June). Simulated brightness-weighted-temperature swaths from an imager onboard a satellite. CEDAR 2023 Workshop, Poster Session, San Diego, CA.

Aguilar Guerrero, J., & Snively, J. B. (2024, June). Synthetic Remote Sensing and Modeling Approaches for Studying Atmospheric Gravity Waves Across the Mesosphere-Lower-Thermosphere Region. CEDAR 2024 Workshop, Poster Session, San Diego, CA.

Aguilar Guerrero, J., Heale, C. J., & Snively, J. B. (2022, December). Characterization of line-of-sight effects for imagers using a periodogram-based analysis. In AGU Fall Meeting Abstracts (Vol. 2022, pp. A13D-04).

Aguilar Guerrero, J., Heale, C. J., Inchin, P., Snively, J. B., & Zettergren, M. D. (2022, December). Visualization of volumetric atmospheric data using modern graphic engines and augmented reality. In AGU Fall Meeting Abstracts (Vol. 2022, pp. SH55B-02). *Invited*.

Kjellstrand, C. B., Bossert, K., Inchin, P., Eckermann, S. D., & Aguilar Guerrero, J. (2022, December). Quasi-two-day wave influences on wave ducting and ionospheric disturbances over the Andes Lidar Observatory. In AGU Fall Meeting Abstracts (Vol. 2022, pp. A15I-1348).

Aguilar Guerrero, J., C. J. Heale, P. A. Inchin, J. B. Snively, M. D. Zettergren, Visualization and characterization of 3D structures of GWs produced by different sources using simulated data, AGU Fall Meeting, SA008-0022, 2020.